

Motion graphs — recall + jargon

Companion to the approved 15 July 2026 Lampros lesson guide. Say the method aloud before checking each answer.

Evidence boundary: Q001–Q011 only. Q012 is deliberately not reconstructed. Q011's later developed solution is labelled transcript-supported / derived; this bundle makes no visual claim after 00:42:30.

30-second graph dictionary

- s–t gradient = v. A horizontal s–t line means rest.
- v–t gradient = a; signed v–t area = displacement. Use absolute areas for distance.
- a–t area = Δv ; recover velocity by adding the initial velocity.
- Velocity and acceleration with the same sign: speeding up. Opposite signs: slowing down.
- Read axes, units, signs, initial condition and modelling words before calculation.

Question-method recall

ID	METHOD	CHECK
Q001	Distance versus displacement	Only for direct one-direction travel from start to finish without reversing.
Q002	Return to the start	Displacement = 0; distance is non-zero (twice the outward length if it retraces the route).
Q003	a–t area and reversal	Use $v(t)=u+ \int a \, dt$. Reversal needs cumulative signed acceleration area to reach and cross $+5 \, \text{m s}^{-1}$.
Q004	Area under a non-negative v–t graph	60 m (option C).
Q005	Bounce graph: apex versus contact	C is the first post-bounce apex. A is $v=0$ during floor contact, not that apex.
Q006	Maximum velocity on s–t	B: it has the steepest positive tangent.
Q007	Terminal velocity and drag	At release $a \approx g$ because drag=0. At terminal velocity, $\Sigma F=0$ because drag equals weight; neither force is zero.
Q008	What an accelerometer indicates	A: level water surface, because acceleration is zero.
Q009	Parachutist acceleration	Initially constant downward g; a large acceleration opposite the downward motion on opening; then acceleration tends to zero at the new terminal speed.
Q010	Convert s–t to v–t and a–t	v–t: constant positive, then 0, then a straight line starting at $v=0$ and decreasing negative. a–t: 0 within the first two open intervals, then constant negative.
Q011	Bouncing ball: v–t to s–t and a–t	s–t has parabolic flight arcs with lower successive rebound apexes; a–t is constant gravitational acceleration between impacts, with idealised positive impulse-like spikes only if impacts are represented.

Jargon to explain accurately

Displacement

MEANING

Signed change in position from start to finish.

AVOID

It can be zero after a non-zero journey.

Distance

MEANING

Total path length travelled.

AVOID

It is never reduced by a return journey.

Signed area

MEANING

Area below a time axis counts as negative for displacement or Δv .

AVOID

Distance from v-t uses absolute areas instead.

Tangent gradient

MEANING

The instantaneous slope of a curve. On s-t it is velocity.

AVOID

s/t is generally an average, not the tangent gradient.

Turning point

MEANING

A point where instantaneous velocity is zero.

AVOID

Its physical meaning depends on the surrounding motion.

Terminal velocity

MEANING

Constant velocity reached when net force is zero.

AVOID

Drag and weight are equal, not both zero.

Accelerometer

MEANING

A device responding to acceleration.

AVOID

It does not directly report the sign of velocity.

Ideal discontinuity

MEANING

An instantaneous jump in a modelled quantity.

AVOID

It needs an impulse/undefined-at-the-instant caveat, not a finite ordinary value.

Parabolic s-t

MEANING

For uniform acceleration, displacement has quadratic time dependence.

AVOID

Infer this only when stated/known, not from casual curve appearance.

Transcript-supported / derived

MEANING

A conclusion grounded in spoken lesson content and standard graph relations where later board visuals are unavailable.

AVOID

It is not a claim about unrecoverable visual geometry.

Focused redo actions

1. Redo Q003 by writing $v = u + \text{signed } a-t \text{ area}$ at each stage; mark exactly where v reaches zero.
2. Redo Q004 with one positive and one negative $v-t$ region: report net displacement and distance separately.
3. Redo Q005 by labelling “contact zero” versus “free-flight apex zero.”
4. Redo Q007/Q009 with free-body arrows at release and terminal velocity.
5. Redo Q010 segment-by-segment, including the ideal discontinuity caveat.
6. Redo Q011 from the given $v-t$ invariants only; do not invent unavailable visual scale or geometry.

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